Yulong Liu

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Research Interests

AI methods & discovery: Differentiable programming, hybrid neural PDE solvers, uncertainty-aware scientific AI, and AI for Science and scientific discovery.

Engineering: Homogenization and multiscale modeling, coupled multiphysics simulation, and subsurface energy systems, including geothermal reservoirs and hydraulic fracturing.

Education

Ph.D. in Earth Science – Cornell University 2024 – Present
Minors: Computer Science and Scientific Computing
B.S. in Mining Engineering – Northeastern University (CN) 2024

Research Experience

Ph.D. Candidate – Dept. of Earth and Atmospheric Sciences, Cornell University Aug. 2024 – Present
Advisor: Chloé Arson *Ithaca, NY*
Project: Scientific AI and scientific discovery for multiscale, multiphysics geomechanics and subsurface energy systems.

- Develop coupled thermo-hydro-mechanical models in MOOSE for subsurface energy systems, including geothermal reservoirs and hydraulic fracturing.
- Study differentiable programming and hybrid neural PDE solvers, combining physics-informed networks, implicit neural representations, and neural operators with numerical simulation.
- Investigate uncertainty-aware surrogate modeling for geothermal systems, with an emphasis on uncertainty propagation and the reliability of multiphysics predictions.
- Lead a cross-institution project on PDE-grounded verification of the physical correctness of LLM-generated multiphysics simulation code.
- Study homogenization and multiscale transport in heterogeneous porous media; use AI for scientific discovery to learn interpretable effective permeability laws.

Undergraduate Researcher – Northeastern University (CN) Sep. 2020 – Jun. 2024
B.S. in Mining Engineering *Shenyang, China*
Thesis: A Joint Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning

- Developed a joint deep-learning model for microseismic signal classification and arrival-time picking, integrating tasks typically handled in separate workflows.
- Conducted experimental and computational research on hydraulic fracturing, triaxial rock testing, and tunnel boring machines; received the Outstanding Thesis Award and National Scholarship.

Grants, Proposals, and Computational Awards

Homogenization and Machine Learning for Multiscale Porous Media Flow 2026
NSF ACCESS project (EES260208). PIs: ***Y. Liu*** *and C. Arson.*

Learned Uncertainty-Propagation and World Models for Enhanced Geothermal Reservoirs 2026
NSF ACCESS project (EES260150). PIs: ***Y. Liu*** *and C. Arson.*

AI-aided Computational Geomechanics 2026
Empire AI Alpha+Beta. PIs: Y. C. Han, ***Y. Liu****, A. Tristani, and C. Arson. 5.5K GPU hours.*

NASA FINESST Proposal – Title Withheld During Anonymous Review 2026
NASA ROSES-25 F.5 FINESST. Under review. Future Investigator: ***Y. Liu****; PI and faculty mentor: C. Arson.*

Publications

Selected Papers

- **Y. Liu**, J. Botvinick-Greenhouse, Y. Yang, C. Arson. *Operator Learning Surrogate Modeling of Hydraulically Fractured Geothermal Injection–Production Systems: A Cornell Case Study.* ARMA US Rock Mechanics/Geomechanics Symposium, Tucson, AZ, 2026.
- **Y. Liu**, C. Arson. *A Physics-Informed Neural Network for Pressurized Cavities of Arbitrary Shape in Heterogeneous Rock. Rock Mechanics and Rock Engineering* (2026). doi:10.1007/s00603-026-05882-5.
- M. Belachew, **Y. Liu**, J. D. Frost, C. Arson. *Numerical Assessment of Plasticity Development and Energy Expenditure of Ant-Like Microtunnelling. Tunnelling and Underground Space Technology*, 172, 107501, 2026.
- **Y. Liu**[†], C. Arson. *Physics-Informed Neural Network Surrogate Modeling of Pressurized Cavity in Homogeneous and Bilayered Media.* ARMA US Rock Mechanics/Geomechanics Symposium, D022S018R006, 2025.
- **Y. Liu**[*,†], Z. Song[*], et al. *Your Simulation Runs but Solves the Wrong Physics: PDE-Grounded Intent Verification for LLM-Generated Multiphysics Simulation Code.* arXiv:2605.09360. Under review at NeurIPS 2026.
- Z. Song, Y. Li, **Y. Liu**[†]. *Prefix-Safe Bayesian Belief Tracking for LLM Reasoning Reliability: Separating Calibration from Ranking.* arXiv:2605.27712. Under review at ACL ARR 2026.

Thesis

- **Y. Liu**. *A Joint Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning.* Undergraduate thesis, Northeastern University, 2024.

[*] Co-first author [†] Corresponding author

Conferences and Presentations

Invited participant, CHiMaD Phase Field Methods XXI — Upcoming
Center for Hierarchical Materials Design (CHiMaD)

Poster presentation, 60th US Rock Mechanics/Geomechanics Symposium (ARMA) — Jun. 2026
American Rock Mechanics Association, Tucson, AZ, USA

Panelist, CHiMaD Phase Field Methods XX — May 2026
Center for Hierarchical Materials Design, Northwestern University, Evanston, IL, USA

Poster presentation, Cornell CEE Graduate Research Symposium — Apr. 2026
Cornell University, Ithaca, NY, USA

Invited seminar: "Physics-Informed AI Optimization of Heterogeneous Rock Systems" — Dec. 2025
Southwest Petroleum University seminar series, Chengdu, China

Poster presentation, 59th US Rock Mechanics/Geomechanics Symposium (ARMA) — Jun. 2025
American Rock Mechanics Association, Santa Fe, NM, USA

Fellowships and Awards

Estwing Hammer Award – Most Outstanding EAS Graduate Student of 2024, Cornell University — 2025
Cornell University Travel Grant, Cornell University — 2025, 2026
Outstanding Thesis, Northeastern University — 2024
National Scholarship, Ministry of Education, China — 2024
First Class Scholarship, Northeastern University — 2023
Autumn Scholarship, Northeastern University — 2022
Golden Seed Scholarship, Northeastern University — 2022

Academic Service

- Session Chair, GS03 Computational Geomechanics, Coupled Thermo-Hydro-Mechanical-Chemical (THMC) Processes, and Next-Generation Modeling, Geothermal Track, *61st U.S. Rock Mechanics/Geomechanics Symposium (ARMA 2027)*, Austin, TX, June 2027
- Reviewer, *Rock Mechanics and Rock Engineering*
- Reviewer, *Underground Space*
- Reviewer, *International Conference on Artificial Neural Networks*
- Reviewer, *Geothermal Rising Conference*

Skills

Programming & Scientific Computing: Python, C++, R, MOOSE finite element framework

Modeling & Simulation: finite element modeling, coupled multiphysics simulation, poromechanics, computational geomechanics, CAD (e.g., AutoCAD, SolidWorks)

Subsurface & Reservoir: geothermal reservoirs, hydraulic fracturing, injection–production modeling, homogenization and multiscale transport in porous media

Scientific Machine Learning: differentiable programming, hybrid neural PDE solvers, uncertainty-aware scientific AI, AI for Science and scientific discovery; physics-informed networks, neural operators, and implicit neural representations

LLMs for Simulation: using large language models to accelerate simulation-software workflows and to verify the physical correctness of generated simulation code

Experimental Methods: rock mechanics testing (e.g., triaxial compression), laboratory instrumentation

Currently Learning: phase-field modeling (fracture/multiphysics), multiphase flow in porous media

Scientific Writing: LaTeX / Overleaf

Professional Memberships

- Member, American Rock Mechanics Association (ARMA), 2025–Present
- Member, International Society for Rock Mechanics and Rock Engineering (ISRM), 2025–Present
- Member, Chinese Society for Rock Mechanics and Engineering (CSRME), 2024–Present
- Member, Geothermal Rising, 2025–Present